Week 5 · Part E — justification

1 · Which variable got which channel, and why

Ranked by how precisely the reader must compare them (position > length > angle > area > hue; Week 2): **x = GDP per capita** and **y = healthcare access** take position, because the question is their relationship and it needs the best channel. **Population** takes mark area, the fourth-best channel, because it is context, not the question; the size legend carries the real values and area is linear in population (sizes=(12,1500) pt², diameter ratio 11.2x, not radius). **Region** takes hue plus marker shape because it is identity; six hues is the most I would use, and the worst pair under deuteranopia is dE 7.5, so shape and direct labels are the second channel. cases_per_100k, the fifth variable, goes to the parallel-coordinates figure.

2 · The parallel coordinates decisions

Scaling: percentile rank, not min–max, because one country's GDP is 9.2x the second and min–max leaves 58 of 60 below 0.10; rank spreads it (Part B.3). z-score would leave Aldonia at +7.5 sd. **Axis order: greedy |r| chain** (healthcare_access → cases_per_100k → urban_pct → median_age → gdp_pc), computed with geolib.corr_chain before looking. **Brushing:** grey all, orange the Borealis region, the one with the highest mean case rate (115/100k), a rule fixed in advance. A scatter matrix shows ten separate pair plots; the parallel plot shows what a single group does on every axis at once: Borealis sits low on access and urban share and spikes on cases in one picture.

3 · The spatial decisions

Denominator: resident population, so the legend reads cases per 100,000 residents; the question is risk, and a count tracks people (r = +0.986). **Classification:** Jenks natural breaks, 5 classes (GVF 0.969, equal to equal-interval and better than quantile at 0.908); with no outliers the natural breaks are honest, and the breaks are printed. **Projection:** Lambert cylindrical equal-area with one reference latitude for every layer, because colour fills area, so area is the encoding channel; Mercator would inflate the federation by 1.34x (1.24x south to 1.49x north). Join: 24 matched, 0 failed. Nothing was suppressed: the smallest district has 468,000 people, so a Poisson check gives no reason to merge. Boundaries are the supplied synthetic file; no source year is stated in it.

4 · The counterfactual

A raw-count map would have darkened D17, D13 and D09 (5,588, 4,048 and 3,437 cases), the three biggest districts, and named them the outbreak; their rates are 51, 64 and 50 per 100,000 against a national 52.8, so two of them are below average. The rate map points instead at D19, D20 and D21 (76, 74, 69), together 4.2 million people who would have had no share of the attention (overlap between the two lists: zero). The five most populous districts hold 59.9% of the people on 12.6% of the land, so a count map is a population map. **Reader evidence:** the two-map activity in Part C has *not* been run on real readers yet; partC_reader_results.csv does not exist and I have made no claim about how readers behave. Ranking-wise, the two lists have no county in common (Dunmere, Pellbrook, Grimsby by count; Hollowfield, Netherby, Elmsworth by rate; r = +0.935), so the same question about which county needs police would produce different answers from the two maps.